



CLOUD NETWORKING THROUGH THE OSI ARCHITECTURE

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INTRODUCTION

- Cloud networking is the connection between users and cloud services over the internet. It allows people to access applications, files, and data from anywhere in the world.
- The OSI (Open Systems Interconnection) model is a framework with 7 layers that explains how data travels from one device to another.
- Each layer has a specific job, making communication secure and reliable.
- In cloud networking, the OSI model helps us understand how information moves between a user's device and a cloud server.



OSI LAYERS IN CLOUD NETWORKING

- Application Layer – User access to cloud applications.
- Presentation Layer – Data encryption and formatting.
- Session Layer – Maintains communication sessions.
- Transport Layer – Ensures reliable data transfer.
- Network Layer – Routing of cloud traffic.
- Data Link Layer – Frame transmission.
- Physical Layer – Hardware and transmission media.



WORKING OF CLOUD NETWORKING

- A user opens a cloud application such as Gmail or Google Drive.
- The request is sent through the seven OSI layers.
- Each layer performs its specific task, such as formatting, routing, and securing the data.
- The data travels through the internet to reach the cloud server.
- The cloud server processes the request and sends a response back to the user.
- The response again passes through all the OSI layers before appearing on the user's device.
- **Simple Flow:**
 - User Device → OSI Layers → Internet → Cloud Server → OSI Layers → User Device



ADVANTAGES AND APPLICATIONS

Advantages:

- Improved scalability
- Better security
- Easy troubleshooting
- Efficient communication

Applications:

- Google Drive
- Microsoft Azure
- Amazon Web Services (AWS)
- Online Banking
- Video Streaming Platforms

CONCLUSION

- The OSI model provides a structured approach to cloud networking.
- It ensures efficient, secure, and reliable communication.
- Understanding OSI layers helps in designing and managing modern cloud systems.
- Cloud networking is a key technology for the future of IT infrastructure.



THANK YOU